PROJECT REPORT

Schrödinger Equation Visualizer

(Particle in a One-Dimensional Box)

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1. Introduction

Computational chemistry combines principles of chemistry, physics, and computer science to solve chemical problems using numerical methods. One of the fundamental concepts in quantum chemistry is the behavior of particles at microscopic scales, which is described by the Schrödinger equation.

This project focuses on visualizing the wave function and probability density of a particle confined in a one-dimensional box using Python. The visualization helps in understanding abstract quantum mechanical concepts in a more intuitive and graphical manner.

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2. Objective

- To solve the time-independent Schrödinger equation for a particle in a 1D box
- To compute the wave function $\psi(x)$
- To compute the probability density $\psi^2(x)$
- To visualize quantum mechanical behavior using graphs
- To understand the concept of quantization and probability distribution

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3. Theory

3.1 Schrödinger Equation

The time-independent Schrödinger equation is given by:

$H\psi = E\psi$

For a particle confined in a one-dimensional box of length L, the allowed solutions are:

$\psi_n(x) = \sqrt{(2/L)} \sin(n\pi x/L)$

Where:

- n = quantum number (n = 1, 2, 3, ...)
- L = length of the box
- x = position of the particle

3.2 Boundary Conditions

- $\psi(0) = 0$
- $\psi(L) = 0$

These conditions imply that the particle cannot exist at the boundaries of the box.

3.3 Probability Density

The probability of finding a particle at a position x is given by:

$P(x) = \psi^2(x)$

This indicates that the particle does not have a definite position but rather a probability distribution.

3.4 Energy Quantization

The energy levels of the particle are quantized and given by:

$E_n = n^2h^2 / 8mL^2$

This shows that only specific energy values are allowed.

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4. Methodology

The following steps were followed to implement the project:

1. Define the input parameters:
    - Length of the box (L)
    - Quantum number (n)
2. Generate a range of x values between 0 and L
3. Compute the wave function using the given formula
4. Calculate the probability density by squaring the wave function
5. Plot graphs using Python libraries
6. Save the generated graphs for visualization

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5. Tools and Technologies Used

- Python
- NumPy (for numerical computations)
- Matplotlib (for plotting graphs)

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6. Implementation

The program is implemented using Python. It takes user input for the length of the box and quantum number, computes the wave function and probability density, and plots the graphs.

The results are displayed graphically and saved as image files.

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7. Results and Observations

The following observations were made:

- The wave function always starts and ends at zero
- The number of peaks in the wave function equals the quantum number (n)
- The number of nodes is equal to (n − 1)
- The probability density is always positive
- As n increases, the number of oscillations increases

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8. Applications

- Understanding quantum mechanical systems
- Visualization of wave functions
- Educational purposes in chemistry and physics
- Basis for advanced simulations in quantum chemistry

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9. Advantages

- Simple and easy to understand
- Provides visual interpretation of abstract concepts
- Requires minimal computational resources
- Can be extended to more complex systems

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10. Limitations

- Limited to one-dimensional systems
- Does not include time-dependent behavior
- Assumes ideal conditions (infinite potential walls)

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11. Future Scope

- Extend to 2D and 3D systems
- Add graphical user interface (GUI)
- Include energy level visualization
- Simulate other quantum systems
- Add animation of wave functions

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12. Conclusion

This project successfully demonstrates the behavior of a particle in a one-dimensional box using computational methods. It highlights the fundamental principles of quantum mechanics such as wave functions, probability density, and quantization of energy.

The graphical representation makes it easier to understand concepts that are otherwise abstract and mathematical in nature.

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13. References

- Quantum Chemistry textbooks
- Lecture notes and class materials
- Python documentation (NumPy, Matplotlib)